

EDUCATION

- MIT** Cambridge, MA
Undergraduate, Computer Science and Mathematics (Double Major) Sep 2024 – Present
Selected Coursework: Discrete Probability and Stochastic Processes; Deep Generative Models; Quantitative Methods for NLP; Representation, Inference and Reasoning in AI; GPA:5.0/5.0
- Tsinghua University** Beijing, China
Preparatory Program, Institute of Interdisciplinary Information Science (IIIS, Yao Class) Sep 2023 – Jun 2024.
Selected Coursework: Advanced Calculus1; Linear Algebra; Abstract Algebra; Algorithm Design; Introduction to Computer Systems; Introduction to LLM Applications; Deep Learning; Exploration to Scientific Research of Lab; GPA:4.00/4.00

SKILLS

- Programming:** Python(Jax/Pytorch for Deep Learning Models), C, C++
- Language:** English(proficient), Chinese(native)

EXPERIENCE

- Undergraduate Research in Deep Generative Models** MIT
Supervised by Kaiming He Sep 2024 -Present
Research on deep generative models in computer vision, especially diffusion models and flow matching models, and investigate the underlying principles of denoising-based models, and develop new models on TPU machines using Jax.
- Undergraduate Research in Technical Computer Science** Tsinghua University
Supervised by William Kuszmaul Feb 2024 -Present
Research on the theoretical analysis of randomized data structures. Developed a new time complexity bound and design an algorithm for a general framework of a randomized problem. Developed a paper to be submitted to PODS/ITCS. (Joint first author)

AWARDS

- Gold Medal** 2023
64th International Mathematics Olympiad(IMO)
- First Prize** 2021
National Olympiad in Informatics in Provinces(NOIP)

PROJECTS

- Video Style Transfer(Deep Learning Course Project)** Feb 2024 - June 2024
Python, Deep Learning
An automatic pipeline to do image style transfer by user-specified prompts using ControlNet and frame interpolation.
- Speeding Up Diffusion Models with One-step Generators** Sep 2024 - Dec 2024
Python, Deep Learning, Generative Models, Diffusion Models, VAE
In the project, we proposed a new method to speed up the training of diffusion models by using one-step generators. On toy experiments, this reduces NFE by half while maintaining the sample quality. We also wrote a blog post, explaining the motivation of the experiment from a higher perspective.
- Knowledge Database** Feb 2024 - Jun 2024
Python, LLM, Bash, Makefile
In the project, we apply LLMs to answer user questions given a folder containing documents as the context. We developed a tagging system, which make the search efficient even when the number of documents is large. We also support semantic search for multimodal documents, such as images and videos.
- Auto GCP TPU Management** Apr 2025
Python, TPU, GCP, Job Management
An automatic job manager for Google Cloud Computing(GCP) development in TPUs. Including automatic enviroment solver, reapplying resuming jobs that use a preemptible TPU which has been preempted, etc.

- **Is Noise Conditioning Necessary for Denoising Generative Models?**
Zhicheng Jiang, Qiao Sun*, Hanhong Zhao*, Kaiming He*